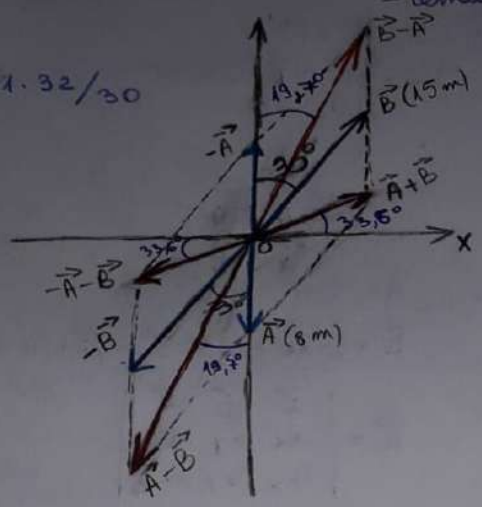
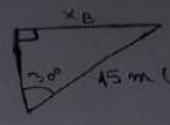
Fizică

– Tema 1 –

1.32/30




$A = (0, -8)$

$B = (15 \sin 30°, 15 \cos 30°)$

$A+B = (15 \sin 30°; 15 \cos 30° - 8) =$

$= (15 \cdot \frac{1}{2}; 15 \cdot \frac{\sqrt{3}}{2} - 8) =$

$\cong (7,5; 4,99)$

$A-B = (-15 \sin 30°, -15 \cos 30° - 8)$

$\cong (-7,5; -20,99)$

$-A-B = -(A+B) \cong (-7,5; -4,99)$

$B-A = -(A-B) \cong (7,5; 20,99)$

Magnitudinea (modulul):

$|A+B| = \sqrt{(7,5)^2 + (4,99)^2} \cong 9,01 \text{ m} \cong |-A-B|$

$|A-B| = \sqrt{(-7,5)^2 + (-20,99)^2} \cong 22,3 \text{ m} \cong |B-A|$

Unghiul (direcția): $\theta = \text{arctg}\left(\frac{y}{x}\right)$

$A+B$: $\theta = \text{arctg} \frac{4,99}{7,5} \cong 33,6°$

$A-B$: $\theta = \text{arctg} \frac{-20,99}{-7,5} \cong 70,3°$ dar ne aflăm în C III $\Rightarrow$

$\Rightarrow \theta \cong 70,3° + 180° = 250,3°$

$-A-B$: $\theta = 180° + \text{arctg} \frac{-4,99}{-7,5} \cong 213,6°$

$B-A$: $\theta = \text{arctg} \frac{20,99}{7,5} \cong 70,3°$

1.42/31

$A = (1,3 \text{ cm}; 2,25 \text{ cm})$

$B = (4,1 \text{ cm}; -3,75 \text{ cm})$

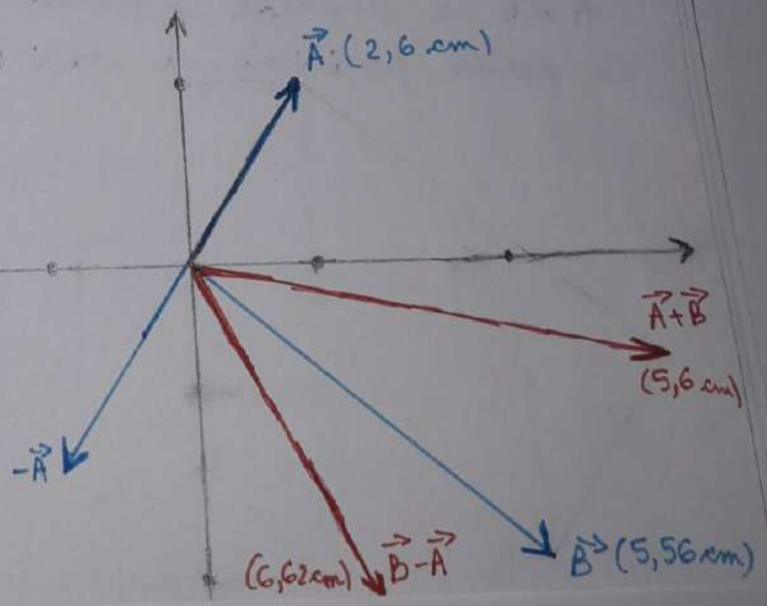
$A+B = (5,4 \text{ cm}, -1,5 \text{ cm})$

$B-A = (2,8 \text{ cm}, -6 \text{ cm})$

$|A+B| = \sqrt{(5,4)^2 + (-1,5)^2} = 5,6 \text{ cm}$

$|B-A| = \sqrt{(2,8)^2 + (-6)^2} = 6,62 \text{ cm}$

(magnitudinea)



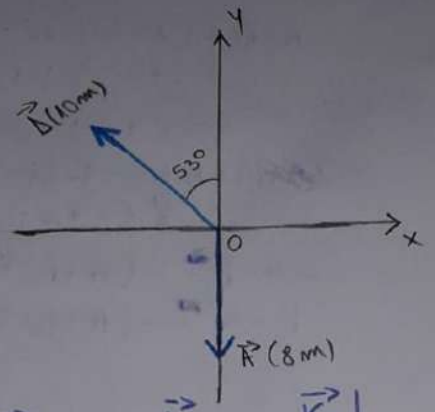
direcția: $A+B$: $\theta = \operatorname{arctg}\frac{-1,5}{5,4} = -15,52°$

suntem în $C_{IV}$ $\Rightarrow \theta = 2\pi + (-15,52°) = 344,48°$

$B-A$: $\theta = \operatorname{arctg}\frac{-6}{2,8} = -64,98°$

suntem în $C_{IV}$ $\Rightarrow \theta = 2\pi + (-64,98°) = 295,02°$

1.57/32




$\vec{A} = (-8\ m)\vec{j}$

$\vec{B} = (-10 \sin 53°\ m)\vec{i} + (10 \cos 53°\ m)\vec{j}$

$$\vec{R} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$\vec{A}\times\vec{B} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -8\ m & 0 \\ -10\sin 53°\ m & 10\cos 53°\ m & 0 \end{vmatrix} = \vec{i}\cdot\begin{vmatrix} -8\ m & 0 \\ 10\cos 53°\ m & 0 \end{vmatrix} + \vec{j}\cdot\begin{vmatrix} 0 & 0 \\ -10\sin 53°\ m & 0 \end{vmatrix} + \vec{k}\cdot\begin{vmatrix} 0 & -8\ m \\ -10\sin 53°\ m & 10\cos 53°\ m \end{vmatrix} =$$

$$= 0\,\vec{i} + 0\,\vec{j} + (80 \sin 53°\ m^2)\vec{k} \cong (63,89\ m^2)\vec{k}$$

$\vec{B}\times\vec{A} = -(\vec{A}\times\vec{B}) = (63,89\ m^2)\vec{k}$

Ambele au magnitudinea $\cong 63,89\ m^2$

$\vec{A}\times\vec{B}$ e orientat în partea negativă a axei Oz, iar $\vec{B}\times\vec{A}$ în partea pozitivă a axei Oz.

2.7/62

$V = \frac{d}{t}$

P-waves : 6,5 km/s

S-waves : 3,5 km/s

$d_p = V_p \cdot t_p$

$d_s = V_s \cdot t_s$

$d_p = d_s - d$

$t_s = t_p + \Delta t$

$\Delta t = 33$ s

↳ delay

$\Rightarrow \frac{V_p/d}{V_s} = \frac{V_s/d}{V_p} + \frac{V_s \cdot V_p}{\Delta t} \Rightarrow d = \frac{\Delta t \, V_s \, V_p}{V_p - V_s} = \frac{11}{25} \cdot 6,5 \cdot 3,5 =$

$= 250,25$ km

$t_p = \frac{250,25}{6,5} = 38,5$ s

$t_s = 38,5 + 33 = 71,5$ s

2.8/62

Honda Civic

$x(t) = \alpha t^2 - \beta t^3$ , $\alpha = 1,5 \, m/s^2$ , $\beta = 0,05 \, m/s^3$

a) $V_{m_1}$? $t_1 = 0$ s; $t_2 = 2$ s

b) $V_{m_2}$? $t_1 = 0$ s; $t_2 = 4$ s

c) $V_{m_3}$? $t_1 = 2$ s; $t_2 = 4$ s

$V_m = \frac{\Delta x}{\Delta t}$ , $\Delta x = x_f - x_i$ , $\Delta t = t_f - t_i$

$x(0) = 0$ m ; $x(2) = 5,6$ m ; $x(4) = 20,8$ m

a) $V_{m_1} = \frac{(5,6 - 0) m}{(2 - 0) s} = 2,8 \, m/s$

b) $V_{m_2} = \frac{(20,8 - 0) m}{(4 - 0) s} = 5,2 \, m/s$

c) $V_{m_3} = \frac{(20,8 - 5,6) m}{(4 - 2) s} = 7,6 \, m/s$

2.20/64

front bumper

$x(t) = 2{,}17\ m + (4{,}8\ m/s^2)t^2 - (0{,}1\ m/s^6)t^6$

a) poziția (x-ul) și accelerația la $V_x = 0$

b) grafice $x-t$; $V_x-t$; $a_x-t$ (sau $(a_x)$) între 0 s și 2 s

a) $V_x(t) = \frac{dx}{dt}$ , $a_x(t) = \frac{dV_x}{dt}$

$V_x(t) = (9{,}6\ m/s^2)t - (0{,}6\ m/s^6)t^5$

b) $a_x(t) = 9{,}6\ m/s^2 - (3\ m/s^6)t^4$

c) $t \cdot V_x = 0$

$V_x(t) = t\left[9{,}6\ m/s^2 - (0{,}6\ m/s^6)t^4\right]$

$\Rightarrow$ $t = 0\ s$ sau $9{,}6\ m/s^2 - (0{,}6\ m/s^6)t^4 = 0$

punem $t = (c)\ s$ , $c \in \mathbb{N}$ (c constant)

$\Rightarrow 9{,}6\ m/s^2 - (0{,}6\ m/s^6)\cdot(c^4)\ s^4 = 0$

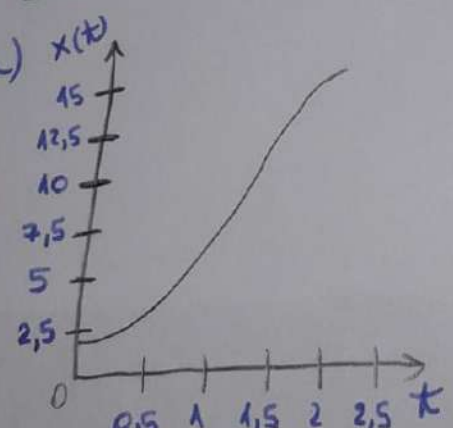
$\Rightarrow c^4 = \frac{9{,}6}{0{,}6} = 16 \Rightarrow c = 2$

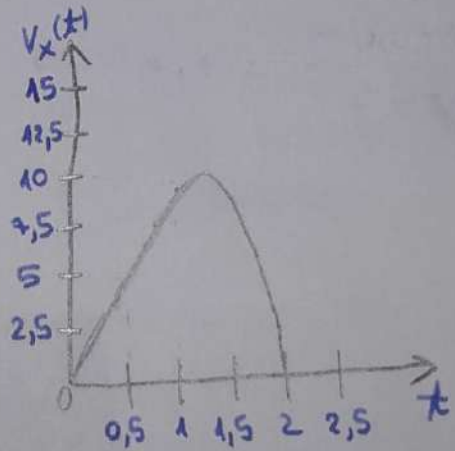
d) deci $V_x = 0$ la $t = 0\ s$ și $t = 2\ s$

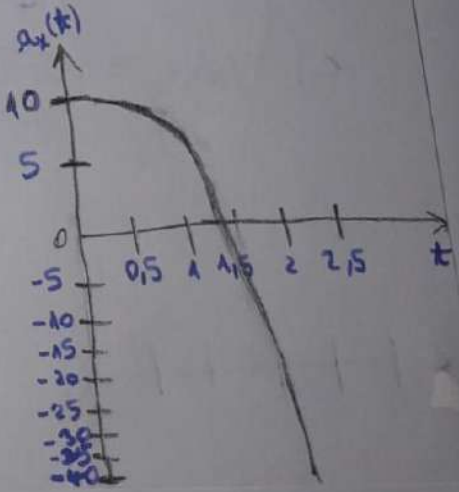
$x(0) = 2{,}17\ m$ , $a(0) = 9{,}6\ m/s^2$

$x(2) = 14{,}97\ m$ , $a(2) = -38{,}4\ m/s^2$

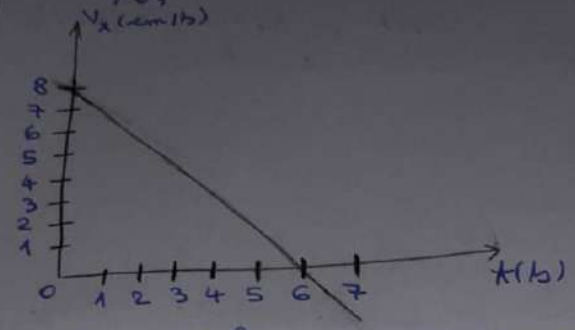
b)



2.29/64




a) $v_{1/2}$? $t_1 = 4s$, $t_2 = 7s$

b) $a_{1/2/3}$? $t_1 = 3s$, $t_2 = 6s$, $t_3 = 7s$

c) $d_{1/2}$? $\Delta t_1 = 4,5s$, $\Delta t_2 = 7,5s$

d) grafic $a_x - x$

a) din grafic $\Rightarrow$ $v(4s) = 2,7$ cm/s

$v(7s) = -1,3$ cm/s

b) $v = v_0 + at$, $a = ct$

$v = 0$ cm/s, $v_0 = 8$ cm/s, $t = 6s$

$\Rightarrow a = \frac{v - v_0}{t} = \frac{-8}{6} = -1,(3)$ m/s$^2$ $\cong -1,3$ m/s$^2$

c) $x = x_0 + v_0 t + \frac{at^2}{2}$

$x - x_0 = d$

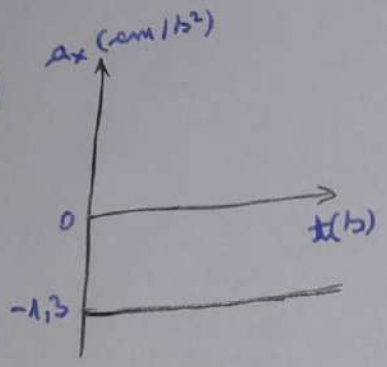
$\forall t \cdot \Delta t = 4,5s \Rightarrow d = 8$ cm/s $\cdot 4,5s + (-1,3$ m/s$^2) \cdot \frac{4,5^2}{2} \cong$

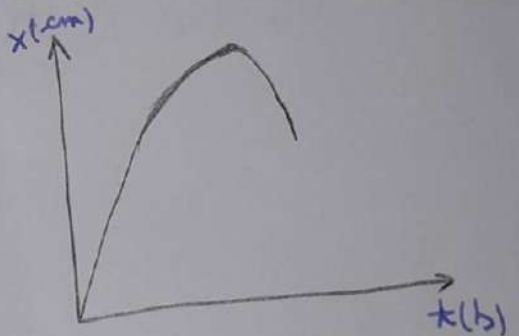
$\cong 22,84$ cm

$\forall t \cdot \Delta t = 7,5s \Rightarrow d = 8$ cm/s $\cdot 7,5s + (-1,3$ m/s$^2) \cdot \frac{7,5^2}{2} \cong$

$\cong 23,44$ cm

d)



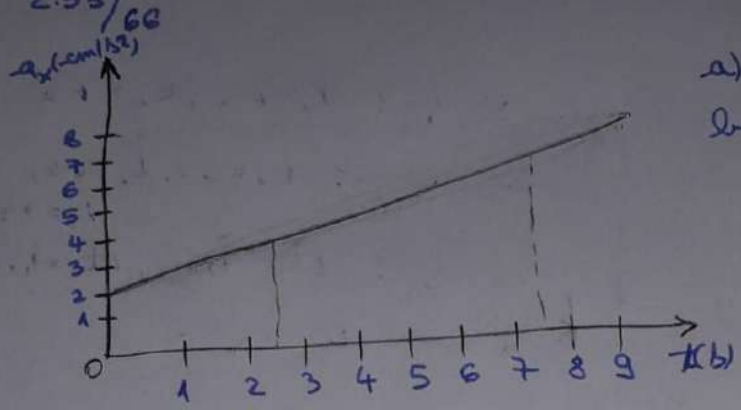




$v_0 = 8$ cm/s

$a = ct = -1,3$ cm/s$^2$

$\Rightarrow$ până la un moment dat crește distanța, după care se micșorează

2.53/66




a) $\Delta V = ?$ $t_1 = 2,5s$ $t_2 = 7,5s$

b) grafic $V(t)$

a) la $t_1 = 2,5s \Rightarrow a_1 = 4 cm/s^2$ (aproximare din grafic)

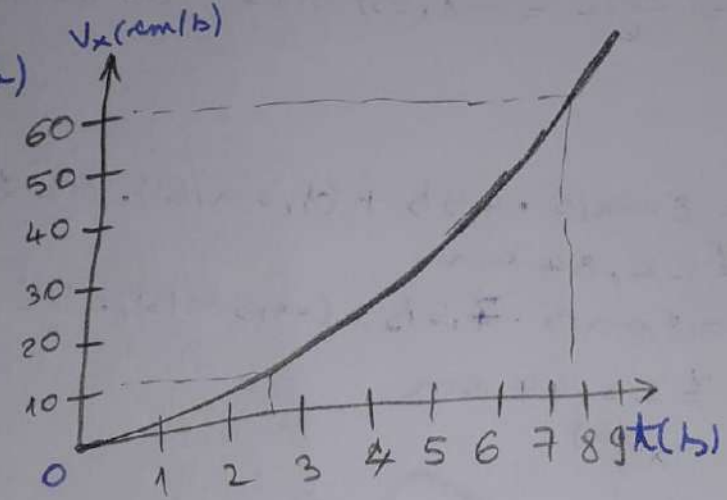
$t_2 = 7,5s \Rightarrow a_2 = 8 cm/s^2$

$\Delta V_x$ = aria subgrafic = $\frac{(a_1 + a_2)}{2}(t_2 - t_1)$

$= \frac{8+4}{2} \cdot (7,5 - 2,5) = 6 \cdot 5 = 30 cm/s$

$= 30$

b)




$V_0 = 0$